

HARIRAMAKRRISHNAN RAMACHANDRAN

+353 89 970 6156 • hariramakrrish@gmail.com • Dublin, Ireland • Stamp 1G — Full-Time Work Eligible

PROFESSIONAL EXPERIENCE

Software Engineer — HCL Technologies, Chennai, India

Sep 2021 – Jan 2025

- Analyzed and mitigated patient safety hazards within **Medtronic software** by collaborating with cross-functional teams, identifying critical risk areas and proposing actionable solutions.
- Conducted comprehensive risk analyses for new product designs, including severity and occurrence assessments and detailed **risk-benefit evaluations** for cardiac medical products.
- Developed and maintained **risk management files** in compliance with ISO 13485, ISO 14971, and IEC 62304, ensuring alignment with design inputs and outputs across multiple product lines.
- Led traceability efforts for **design verification and validation** within risk management documentation, ensuring all safety measures met global regulatory standards.
- Supported and executed reliability and safety engineering studies, including **design of experiments (DOE)**, to evaluate design concepts and system performance for medical devices.
- Recommended and implemented design improvements, test methods, and **statistical process control techniques** to enhance product reliability and patient safety, reducing reported anomalies by **15%**.
- Provided data-driven **risk-benefit insights** and proposed innovative risk mitigation strategies to stakeholders, prioritizing patient safety in all recommendations.

SKILLS

Quality & Safety Standards – ISO 13485, ISO 14971, IEC 62304, PHA, HAL, FTA, DFMEA

Software Development – Java, Python, C++, JavaScript, C#, .NET, Angular

Engineering Principles – Product Reliability, Safety Engineering, Risk Management, Design for Reliability (DfR), Design for Manufacturability (DFM), Design for Six Sigma (DFSS), Robust Design

Methodologies – Agile, Scrum, Design of Experiments (DOE)

Tools & Platforms – Jira, Confluence, Splunk, AWS, Linux

PROJECTS

Medical Device Software Risk Analysis and Mitigation

- Performed detailed hazard analysis and risk assessments for a new implantable device software component, identifying potential failure modes using **FTA and DFMEA** techniques.
- Developed and documented comprehensive **risk management files**, ensuring full compliance with ISO 14971 and IEC 62304 standards for regulatory submission.
- Collaborated with hardware and software engineering teams to implement targeted risk mitigation strategies, leading to a projected **20% reduction** in identified safety concerns.

Reliability Study for Cardiac Product Software

- Designed and executed a **design of experiments (DOE)** to evaluate the reliability of critical software functions under simulated stress conditions.
- Analyzed test results using statistical methods to identify key drivers of performance degradation and formulated recommendations for design improvements.
- Contributed to the development of **statistical process control (SPC)** charts for ongoing monitoring of system performance in production environments.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin

Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India

2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)